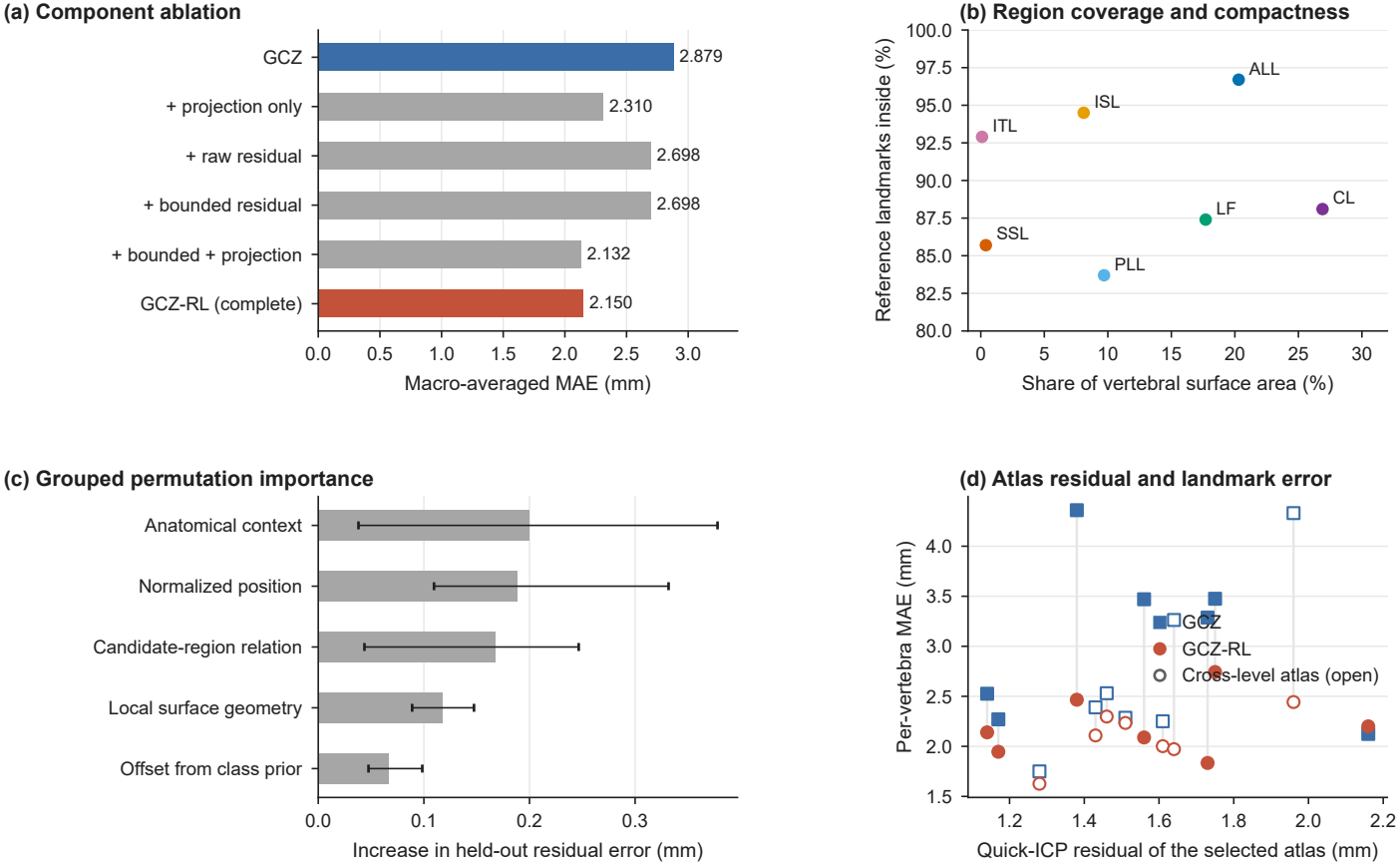

(a) Component ablation
GCZ
+ projection only
+ raw residual
+ bounded residual
+ bounded + projection
GCZ-RL (complete)
2.879
2.310
2.698
2.698
2.132
2.150
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Macro-averaged MAE (mm)
(b) Region coverage and compactness
Reference landmarks inside (%)
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
80.0
ALL
ISL
ITL
LF
CL
SSL
PLL
0
5
10
15
20
25
30
Share of vertebral surface area (%)
(c) Grouped permutation importance
Anatomical context
Normalized position
Candidate-region relation
Local surface geometry
Offset from class prior
0.0
0.1
0.2
0.3
Increase in held-out residual error (mm)
(d) Atlas residual and landmark error
Per-vertebra MAE (mm)
4.0
3.5
3.0
2.5
2.0
1.5
GCZ
GCZ-RL
Cross-level atlas (open)
1.2
1.4
1.6
1.8
2.0
2.2
Quick-ICP residual of the selected atlas (mm)